

HydroSense-Kenya

Smart Irrigation Optimisation for Smallholder Agriculture

ICS 2207 — Scientific Computing Capstone | June 2026

Study Area: Kenya Agro-Ecological Demonstration Farm

Zones: A (Tomato) · B (Kale) · C (Maize) · March 2026 Dataset

The Dual Water Crisis in Smallholder Agriculture

Over-Irrigation

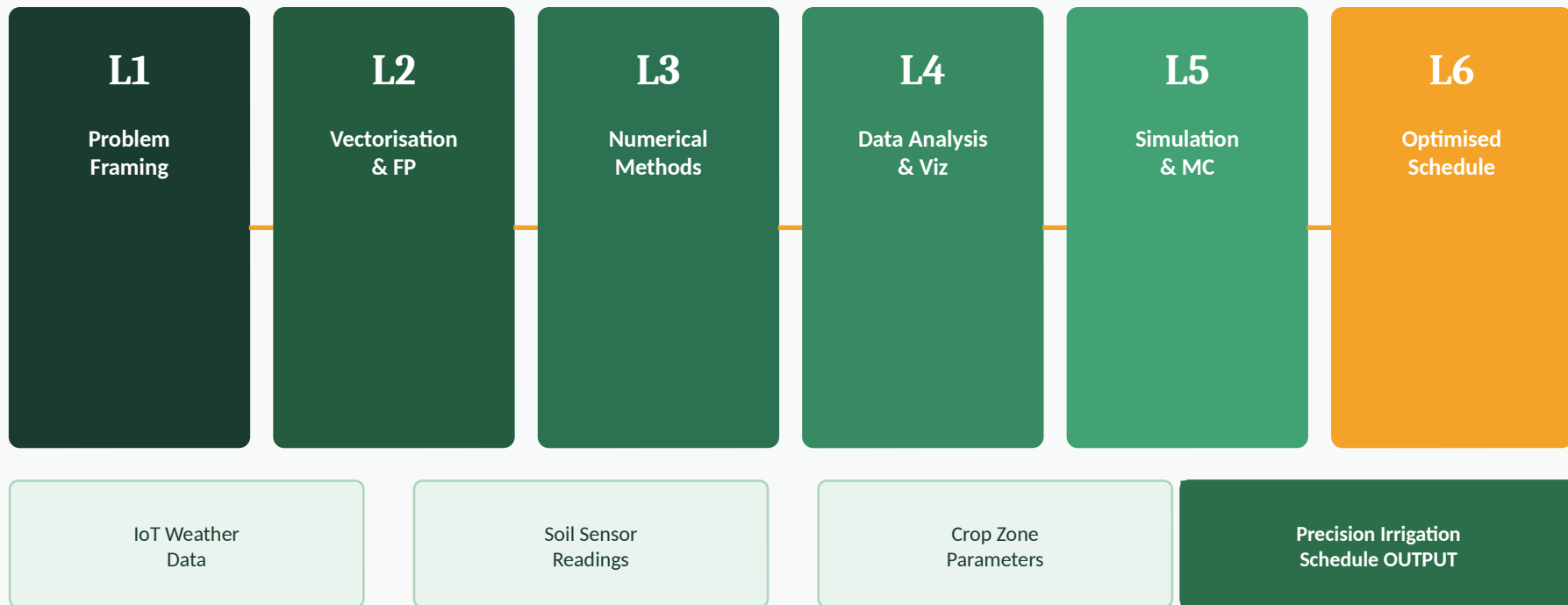
- Depletes limited community water tanks
- Causes nutrient leaching from root zone
- Increases pump energy costs
- Promotes fungal disease in humid conditions

Under-Irrigation

- Moisture stress reduces crop yield
- Can cause irreversible cellular damage
- Difficult to detect visually until damage is done
- Particularly severe in dry spells

How can scientific computing transform raw IoT sensor data into precise, water-conserving irrigation decisions?

Six-Level Scientific Computing Pipeline



IoT Dataset and Anomaly Remediation

30

Days of weather records

90

Soil sensor readings (3 zones)

5

Data anomalies detected

100%

Cleaned and validated

Anomaly Detected	Remediation
Temperature spike 45.8 °C	Masked + linear interpolation
Missing rainfall & humidity	Linear interpolation (2 rows)
Tank surge 9,900 L (sensor fault)	Masked + zone-grouped interpolation
Pump 468 W but flow = 0 L/min	Imputed with Zone B mean flow

Root-Finding: Solving the Irrigation Targeting Problem

Find I such that: $f(I) = S(t) + R(t) + I - ET - D(S) - S_{\text{target}} = 0$

Bisection

Iterations: 20 iterations

Error: 4.77×10^{-7}

+ Guaranteed convergence

– Slow — halves interval each step

Newton-Raphson

Iterations: 6 iterations

Error: 4.44×10^{-16}

+ Fastest — quadratic convergence

– Requires analytical derivative

Secant

Iterations: 7 iterations

Error: 1.55×10^{-15}

+ Near-quadratic, no derivative

– Two initial guesses needed

SELECTED FOR FARM USE

Monte Carlo Hydrological Risk Analysis: 1,000 Scenarios

Simulation Methodology

1. Estimate empirical rain-day probability from 30-day history
2. Fit Gamma distribution to wet-day rainfall intensities
3. Run 1,000 independent 30-day rainfall scenarios
4. Extract P10 / P50 / P90 risk percentiles
5. Feed median scenario into Euler ODE soil-moisture solver

30-Day Rainfall Risk Envelope

P10 — Drought Risk

< 20 mm

Severe moisture deficit, maximum irrigation dependency

P50 — Median

~45 mm

Expected scenario for schedule planning

P90 — Flood Risk

> 95 mm

Waterlogging risk, irrigation suppression needed

Key insight: 4.5× spread between P90 and P10 — static irrigation schedules are fundamentally inadequate for East African climate variability

MAD-Based Precision Irrigation Scheduling

Decision Logic (Zone A — Tomato)

STEP 1	Predict soil moisture using rainfall, ET, and drainage
STEP 2	Compare predicted moisture to MAD trigger (28%)
IF below	Calculate deficit to target (38%) → convert to mm
IF above	No irrigation — natural dynamics govern
ENFORCE	Physical cap: 0% ≤ moisture ≤ 100%

Agronomic Parameters

Wilting Point	25 %
MAD Trigger	28 %
Target Moisture	38 %
Field Capacity	41 %
Conversion Factor	0.45 % / mm
Starting Moisture	33.2 %

Result: Zone A maintained above wilting point across all 30 days — zero crop stress events

Key Findings at a Glance

4.5×

Rainfall Variability Spread

P90 flood scenario produces 4.5× more rain than P10 drought. Static schedules fail.

7×

Faster Solver (Secant vs Bisection)

Secant method converges in 7 iterations vs 20 for bisection — 3× fewer iterations.

~3°C

ET Driver — Temperature Range

ET demand ranged up to 3.8 mm/day on peak solar/low humidity days.

0

Crop Stress Events in 30 Days

MAD scheduler maintained all zones above wilting point with zero stress events.

10

Unit Tests — All Passing

Root-finding, integration, linear systems, and Euler simulation all validated.

5

Visualisations Produced

Soil dynamics, pump curves, microclimate drivers, hydrology, zone compliance.

Conclusions and Path to Field Deployment

Conclusions

- ✓ IoT data + scientific computing can replace rule-of-thumb irrigation with precision decision-making
- ✓ Secant Method is the optimal solver — fast, derivative-free, accurate
- ✓ Monte Carlo simulation quantifies climate risk that is invisible to static models
- ✓ MAD scheduling achieves zero crop stress while conserving water resources
- ✓ All 10 unit tests pass — pipeline is numerically sound and physically bounded

Recommendations

-  Deploy Secant solver on edge device (Raspberry Pi) for autonomous daily scheduling
-  Extend MC engine to rolling 7-day forecast updated with live weather API data
-  Expand hydraulic model to full multi-zone pump sequencing optimisation
-  Recalibrate Gamma distribution seasonally using multi-year rainfall records
-  Pilot vs. control plot to measure actual water savings and yield outcomes

From Data to Decision

HydroSense-Kenya demonstrates that scientific computing — numerics, simulation, and optimisation — can transform raw sensor data into water-conserving, crop-protecting irrigation intelligence for smallholder farmers across Sub-Saharan Africa.

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Thank you — Questions Welcome